

오션테크 코리아

BEYOND NAVIGATION

아비커스 자율운항 상용화 현황과 미래 전략



Avikus
MARITIME AUTONOMOUS PIONEER

December 2023

Dohyeong Lim Ph.D CEO

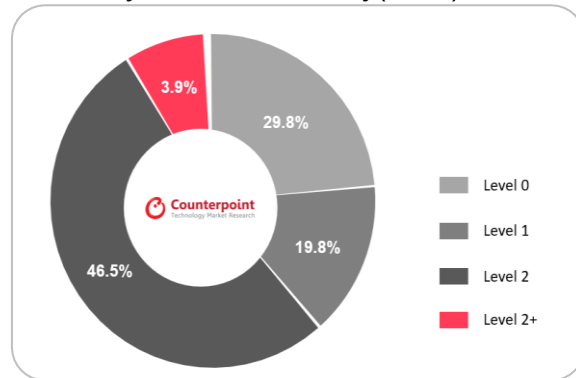
27th Oct. 2023

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Trends in the Autonomous Vehicle Industry

Technologically Feasible, but Commercially Distant

US ADAS sales
by level of autonomy(2022)



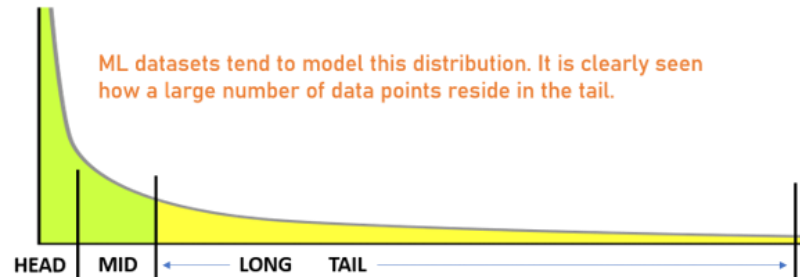
Tesla FSD V12
End-to-end learning



Kia, EV9 Lv.3 'HDP'
Option deletion and refund



Cruiser Robotaxi loses its self-driving
license
due to safety issues

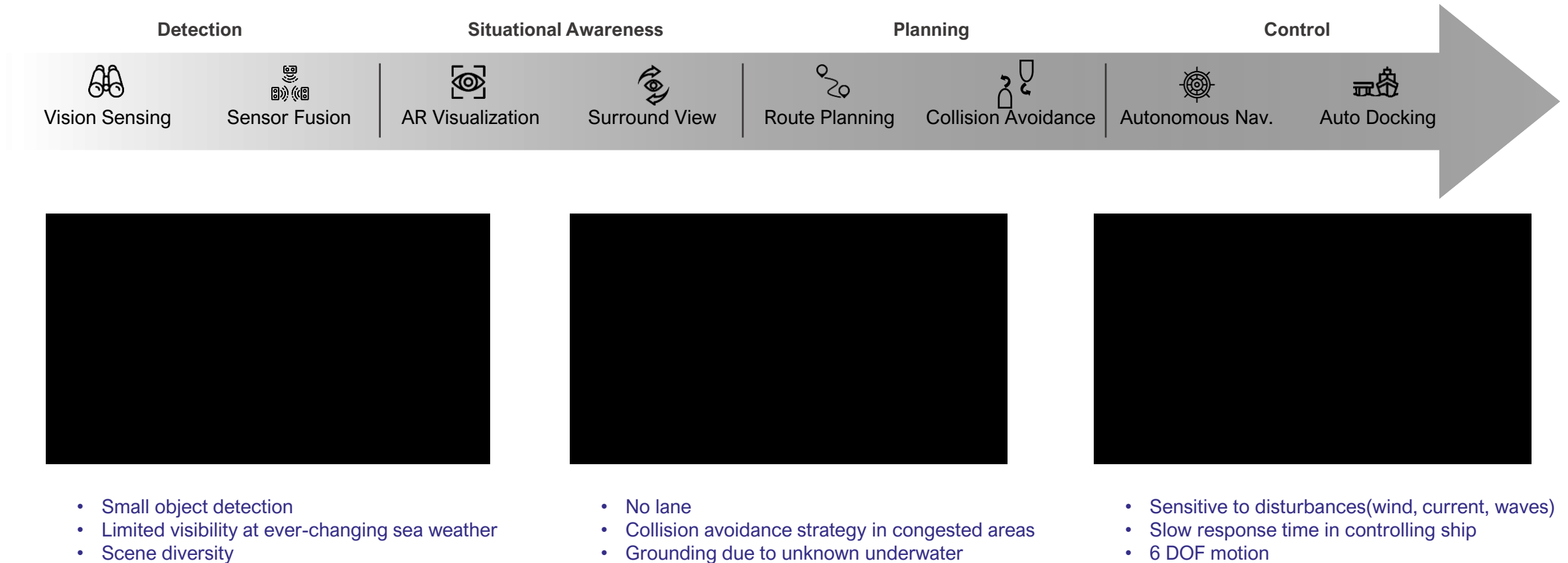


Mature technology, but still a long way to go beyond level 3 (Long tail problem)

→ Similarities with autonomous vessel industry

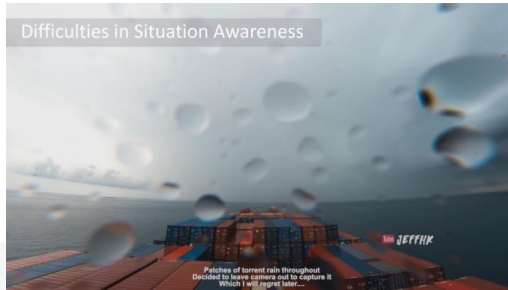
Problem solving for 'autonomous navigation'

Challenges Arising from the Unique Characteristics of Maritime Environments

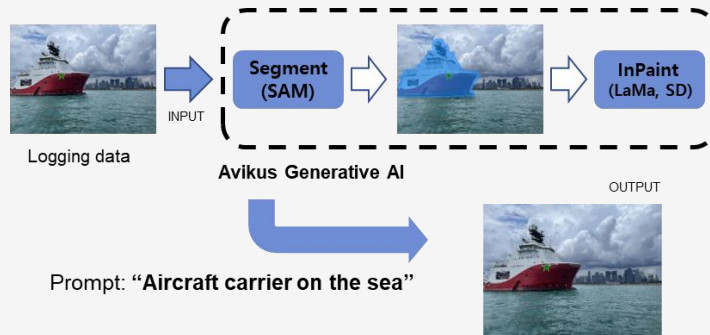


Novel Approach to Tackle These Challenges

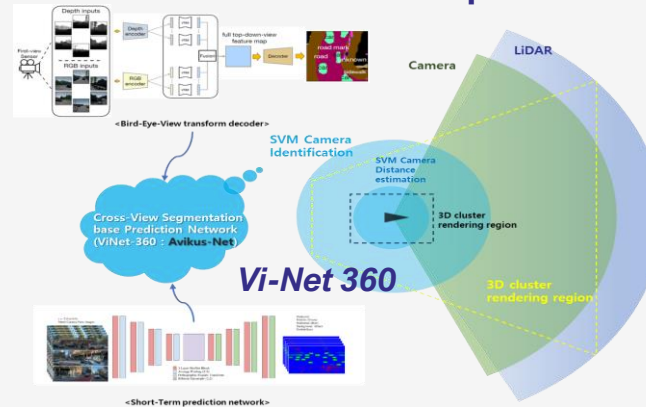
Through Roof Shot, Not Moon Shot



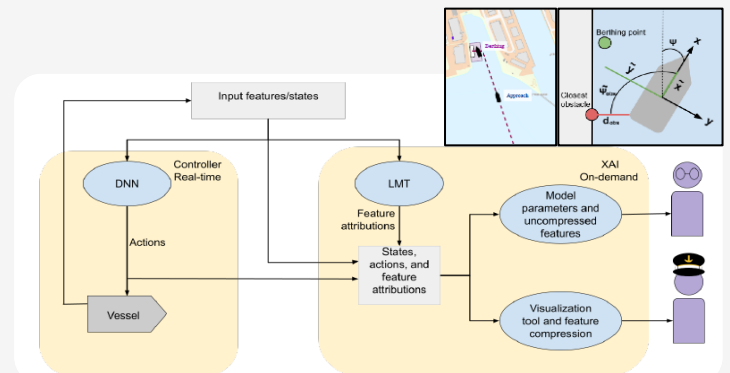
Data generation with generative AI



Custom AI model for 360° prediction



Deep Reinforcement Learning



Business Strategy of Avikus

Considering Technological Advancements and Market Trends

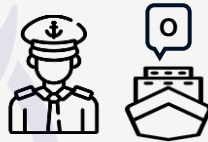
MASS Maritime Autonomous Surface Ship

Current business area of Avikus

NOW



Ship with automated processes and decision support



Remotely controlled ship with seafarers on board



Remotely controlled ship without seafarers on board



Fully autonomous ship

FUTURE

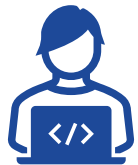
- In compliance with current regulations
- Relatively easier to advance toward commercialization

- Need to consider **product liability**
- May **vary** depending on the concept of the ship

- Uncertain & technically challenging
- Prospect for the **very distant future**

Steadfast Growth of Avikus

A Continuous and Upward Trajectory



90+

Staffing 7 in 2021
Mainly AI S/W engineer



350+

Cumulative orders
100+ orders from non-HD Group



20+

European premium model option
selection after launching 2023



2⁵

Double growth goal
for 5 consecutive years

Stepwise Footprint of Avikus

From the World's First to the World's Best



Elevate the Autonomous Navigation Experience

With HiNAS for Ships and NeuBoat for Boats



Unleash the infinite potential of the ocean by leading the paradigm shift of marine mobility

Safety

- Reduce accidents caused by human error
- Respond to the lack of experienced seafarers

Economy

- Improve fuel efficiency (Just-in-time)
- Reduce operation and new building costs

Sustainability

- Support emission regulations
- Provide relief to seafarers

Safety

- Reduce accidents caused by human error

Convenience

- Make boating easy and accessible to all

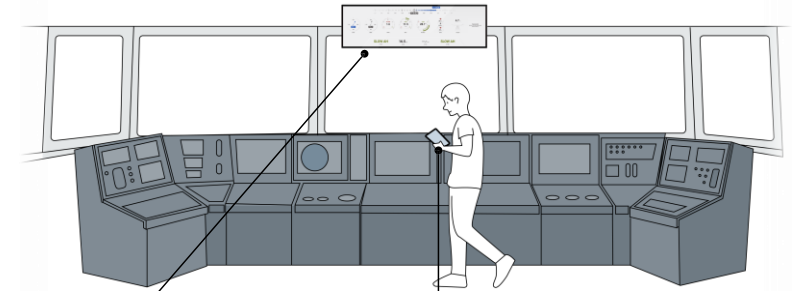
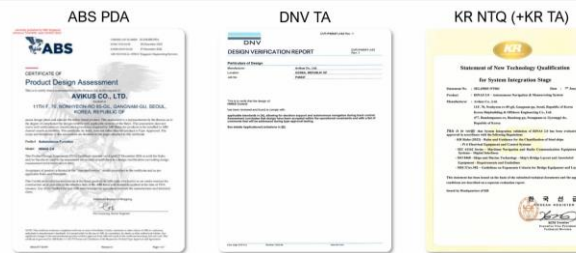
Pleasure

- Reduce driving-related anxiety and maximize carefree leisure time

Avikus HiNAS

“Revolutionizing maritime logistics”

HiNAS is a **partially autonomous navigation system** with functions that include route optimization, autonomous navigation, maneuvering, collision detection, and collision avoidance. Also, it provides OTA(Over-the Air) Software Updates.



Key Features



Situational Awareness



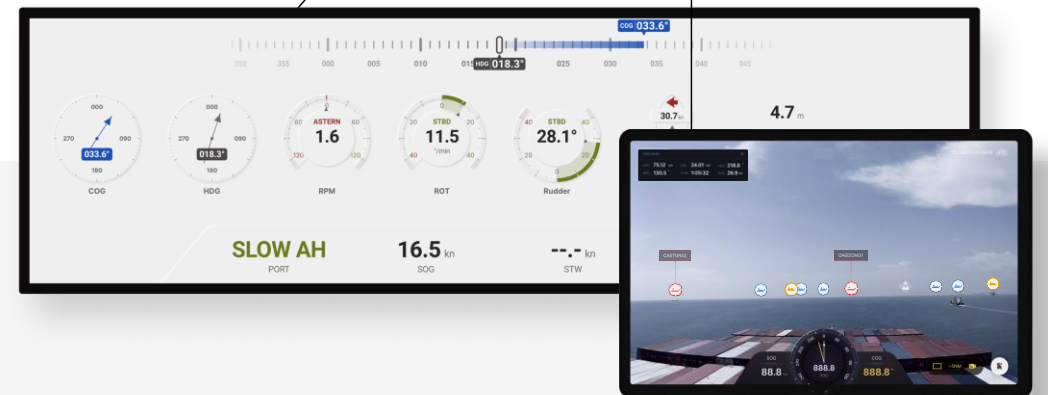
360° Real-Time View



Collision Avoidance Control



Route Optimization & Following



Situational Awareness

Key Feature of HiNAS (1/5)

- Computer vision and deep learning-based **target detection**
- **IR camera-based** target detection at night or restricted visibility
- Using **sensor fusion** technology(RADAR, AIS, Cameras, LiDAR, etc.)



▶ IR(Infrared) Camera (120 deg. FOV)

▶ EO(Electro-Optical) Camera (180 deg. FOV)

Computer vision & deep learning-based objects detections
→ **Algorithm was developed based on more than 5M data points**

vessel(61%)

vessel(90%)

vessel(56%)

vessel(71%)

vessel(37%)

vessel(88%)

Real-Time 360° View

Key Feature of HiNAS (2/5)

- Useful when passing canal, narrow channel, berthing/unberthing
- Prevent the risk of armed robbery, stowaways, smugglers, etc.
- Actual Example from Korean Ship Owner
 - They caught an attempt of a stowaway in advance using HiNAS
- Monitoring from anywhere onboard using the portable device

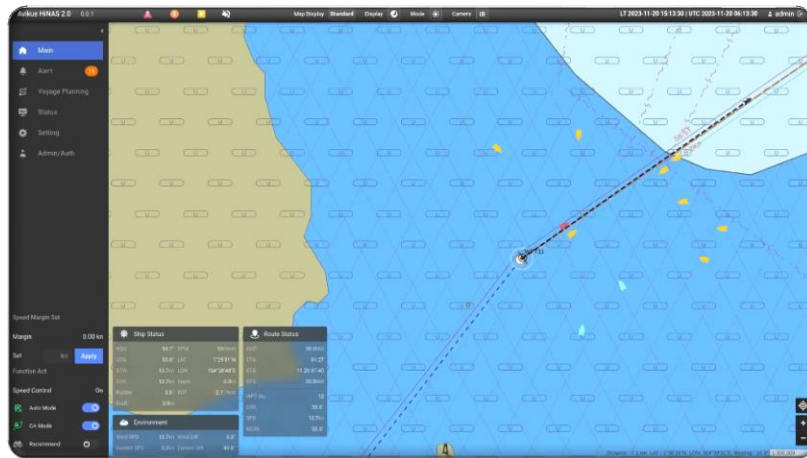
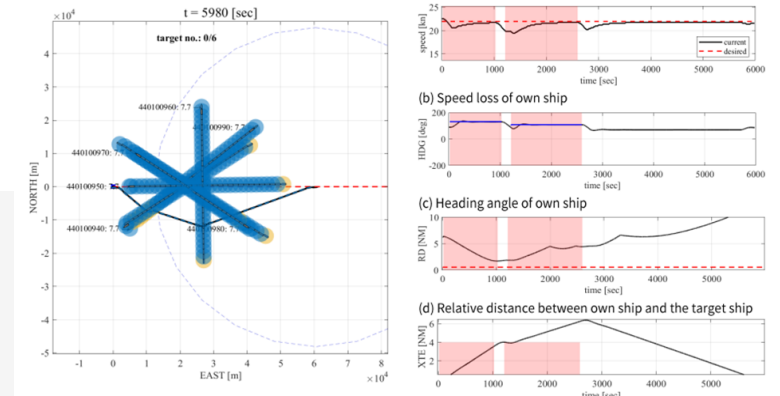


Collision Avoidance Control

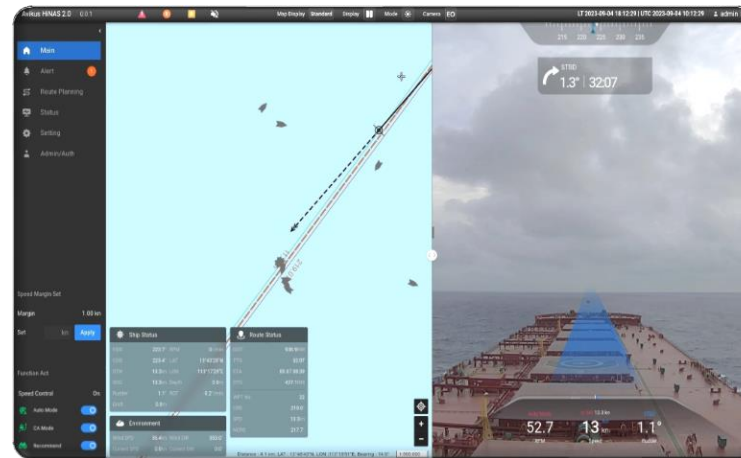
Key Feature of HiNAS (3/5)

- Real-time collision detection and autonomous collision avoidance
- Incorporating experienced seafarers' know-how into algorithms, basically based on COLREG*

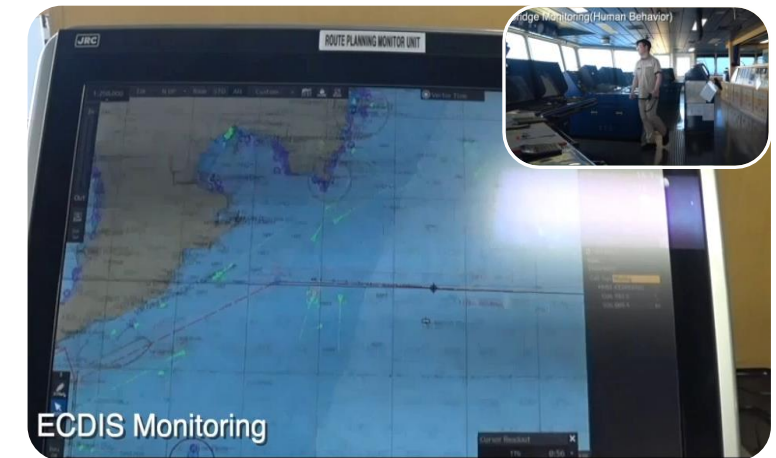
*International Regulations for Preventing Collisions at Sea



HILS(Hardware-in-the-Loop Simulations)



Autonomous transoceanic voyage (actual navigation using HiNAS)

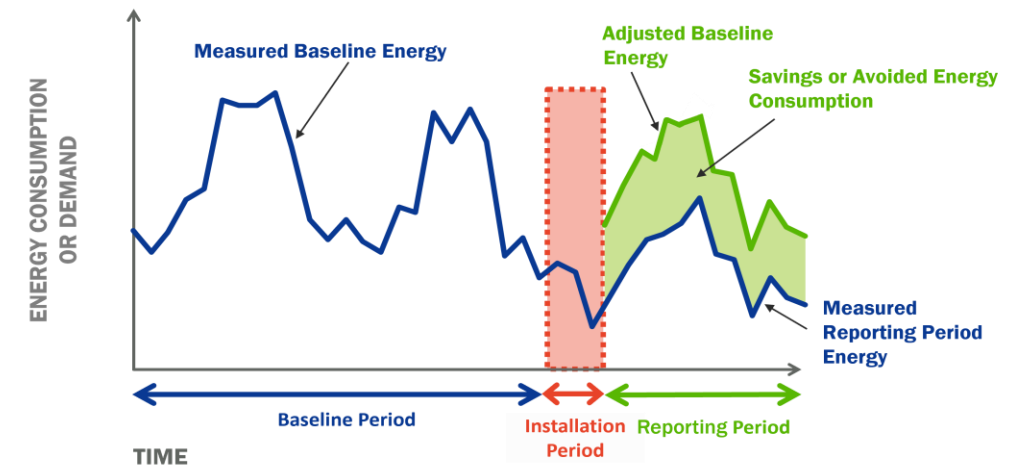
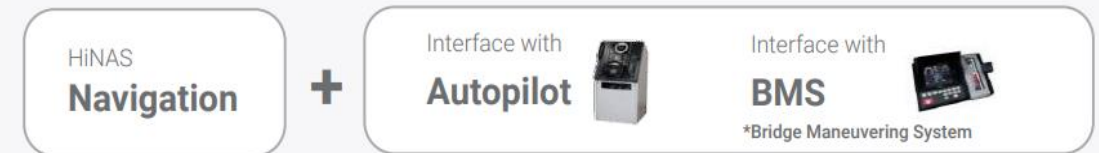
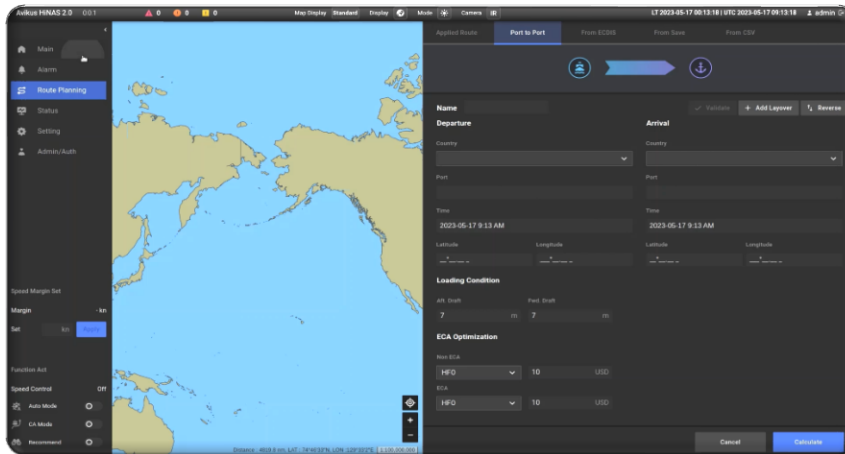


ECDIS Monitoring

Route Optimization & Following

Key Feature of HiNAS (4/5)

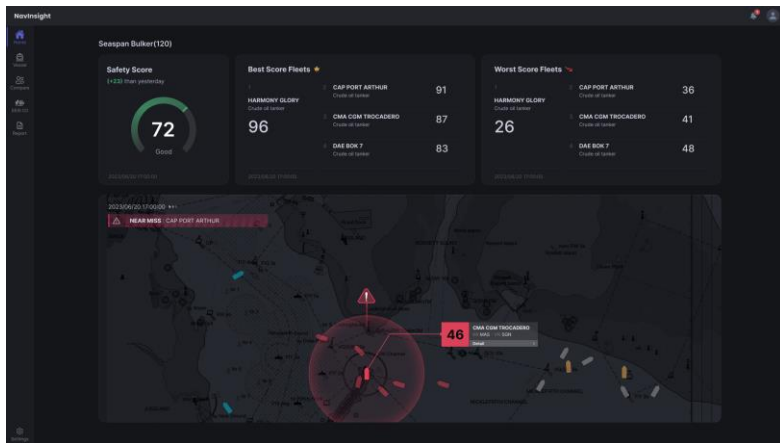
- Executing autonomously through the interface with Autopilot and BMS
- Providing optimal route and speed considering weather information and ship dynamics, etc.
- Increasing fuel efficiency by 7%, while reducing greenhouse gas emissions by 5%
- The most competitive green solution on the market



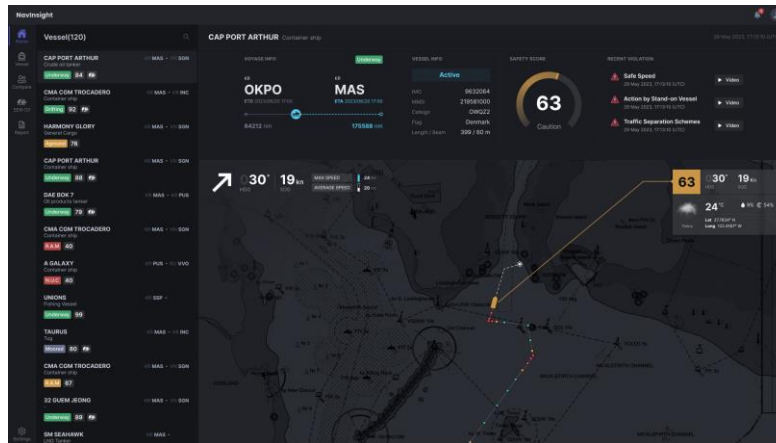
Cloud service

Key Feature of HiNAS (5/5)

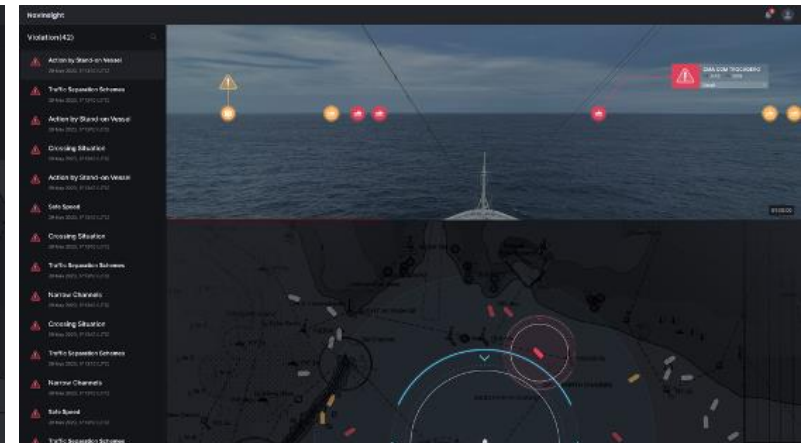
- Providing schedule monitoring, location tracking, OTA software update, black box function, etc.
- Real time fleet monitoring for safety, fuel consumption, and GHG emission.



Fleet Dashboard



Vessel Monitoring



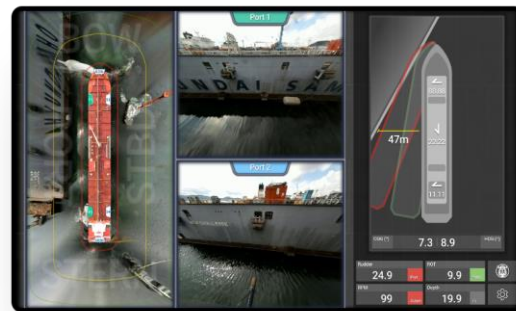
Blackbox Replay

Avikus HiNAS

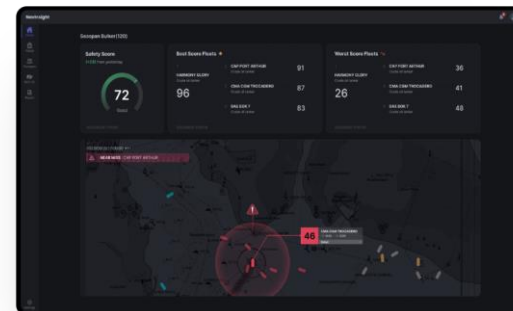
“Revolutionizing maritime logistics”



Navigation



SVM



Cloud



Control

Enhanced Situational Awareness

Autonomous Control

Real Experiences, Real Satisfaction

Unveiling User Testimonials

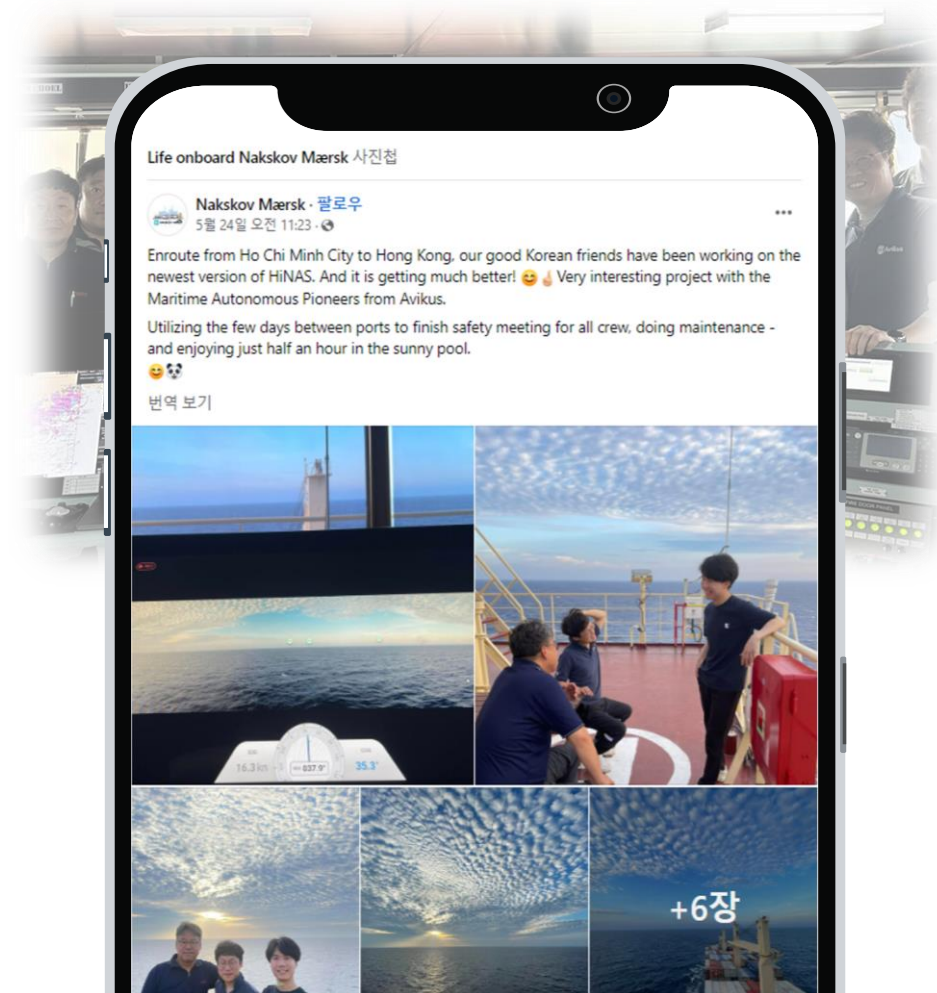


“**FUTURE IS HERE**”

“The new display is **simple yet effectively** present the necessary navigation information, making it very nice!

Especially, it is **intuitive to perceive the risk of collision** using the congested mode and extremely convenient with the addition of wide/full-screen mode.”

Captain of the container ship of Seaspan



Avikus NeuBoat

“Democratizing the boating experience”

NeuBoat is a **partially autonomous navigation** system for recreation boat with functions that situational awareness, auto cruising, auto docking. It is a system that helps users enjoy safe and comfortable boating.



Key Features



Situational Awareness



360° Real-Time View



Collision Avoidance Control



Auto Cruising & Auto Docking



Insights from Users

How Autonomous Navigation Technology Addresses User Inconveniences

“Always keeping watch is stressful”



Object detection & classification

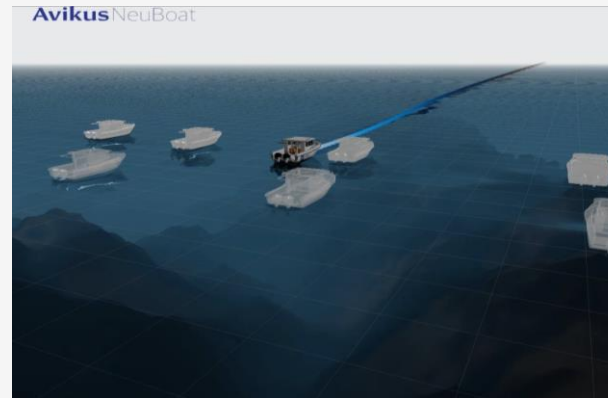


Detection of drowning person

“I want to see blind spots”



3D SVM without blind spot

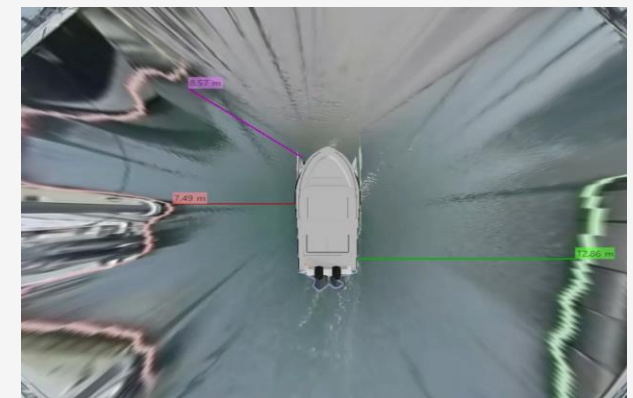


Underwater monitoring & Guide

“Docking is nerve-racking”



Auto-docking control robust to disturbances



Vision only distance estimation

Avikus NeuBoat

“Democratizing the boating experience”



DOCK

360 Birds Eye View



NAVI

Object Detection

Enhanced Situational Awareness



Autonomous Navigation &
Collision Avoidance

CoPilot



Auto-Docking

Autonomous Control

Real Experiences, Real Satisfaction

Unveiling User Testimonials



“

Avikus is breaking down the barriers and ignoring limits!

“Our future as an industry depends on attracting as many people to recreational boating as possible.

Autonomous boating **opens the door to so many prospective boaters and gives added benefits to existing boaters.**

Well done!”

Ken Cliton, President at Intrepid Powerboats & Kim Sweers, Managing Partner of FB Marine Group

Maritime Autonomous Pioneer

Stepwise approach to higher-level of autonomy

2021~ Autonomous Level **01**
Navigation assistant system for ships



AI navigation assistance / Collision avoidance control (HiNAS)



Surround view/Auto berthing (HiBAS)

2023~ Autonomous Level **02**
Autonomous solution for boats



Leisure boat autonomous solution

2025~ Autonomous Level **03**
Autonomous platform as a service



Cloud service platform for ships



Apps for leisure boat autonomous solution

2027~ Autonomous Level **04**
Maritime autonomous mobility business



Marine mobility business based on autonomous navigation

Beyond Navigation

Partner for safer and greener voyage



기존 항해 장비의 한계를 극복하여 상황인지 자동화를 통해 안전을 극대화,
자율 항해를 통한 연비 저감, 탄소 배출량 최소화하는 방향으로 개발
(현존 CAPEX/OPEX 가장 경쟁력 있는 친환경 솔루션)

선박의 자율화/자동화는 시대적 필연이며
단계적/점진적으로 진행될 것임

(그룹사 표준으로 제어 솔루션 탑재 중)

생성 AI 의 적극적인 활용, 최적의 센서 조합 및 AI
모델 선정,
해상 환경에 특화된 제어전략 적용을 통해
충분히 고객에게 가치를 줄 수 있는
ADAS 수준의 자율운항 기술 상용화가 가능하였음



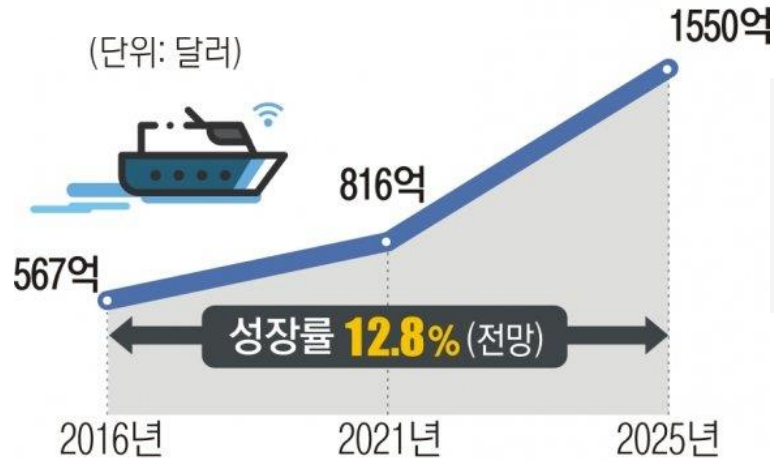
자동차 ADAS와 같이 안전/편의를 제공하여
운전의 진입장벽을 낮추어 보팅을 대중화 하는 방향으로 개발

자율운항 기술은 한국조선/해운/기자재
업계에 큰 기회가 될 것임

(테슬라의 사장가치는 전세계 자동차 OEM
시총 합보다 더 높음)

Beyond Navigation

Partner for safer and greener voyage



자율운항선박 시장 규모

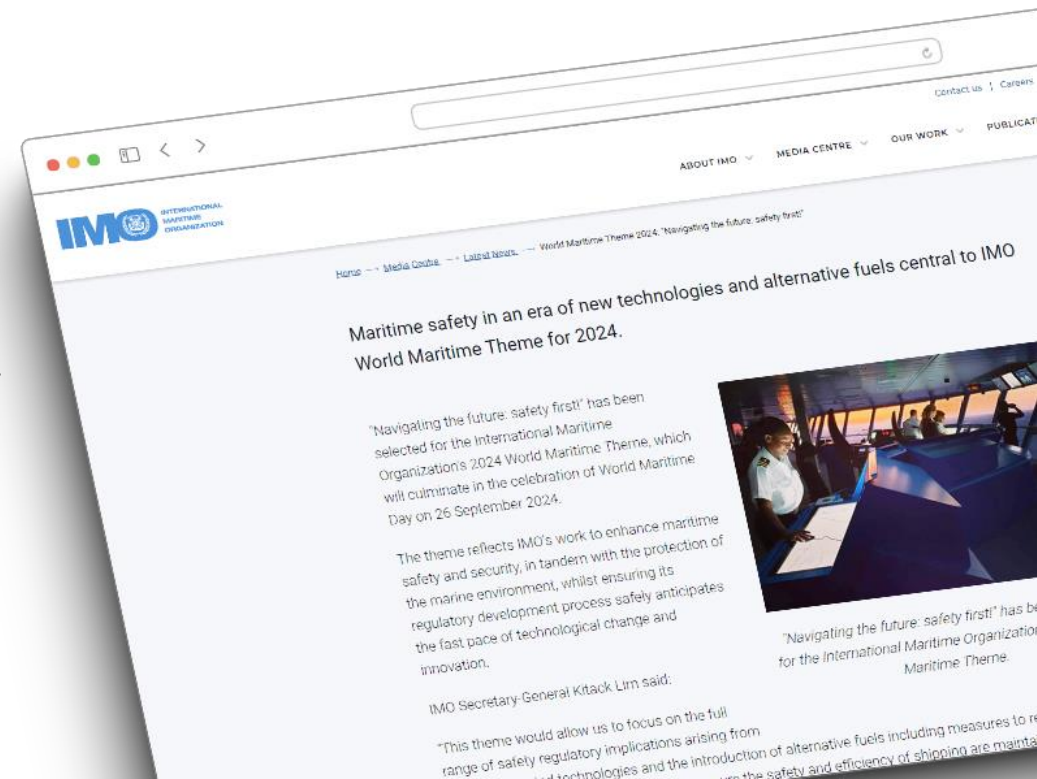
- 기술 성숙도가 확보되고 관련 사회 인프라가 정비된다면 자율운항선박 도입은 가속화될 전망
- 특히 원격제어 혹은 완전자율운항보다 부분자율운항이 96%에 달할 것

* <자료> 해양수산부, Markets and Markets

2028
IMO MASS Code 발효
(강제성을 가짐)

- 국제해사기구(IMO)의 자율운항선박 관련 협약 개정 논의가 본격화
- IMO 는 새로운 기술과 혁신을 수용하는데 필요한 변화를 예측하고 안전을 위한 규제 프레임워크 사전작업을 지속하고 있음
- 그 대표적인 예시가 MASS Code 개발임을 언급함

→ 국제해사 분야에 있어서 현재 가장 큰 영향을 미치는 것이 친환경 규제와 더불어 자율운항선박임을 알 수 있으며 미래를 위한 선제적 대응이 필요함을 의미



Your partner for safer and greener voyage

THANK YOU

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